

Regional distribution and profiling of childhood acute leukemia in Brazil: Report from the EMiLI study group

Maria S. Pombo-de-Oliveira^{1*}, Marcelo G. P. Land², Paulo Chagas Neto³, Daniela P. Mendes-de-Almeida^{1,4}, Isis M. Quezado Magalhães⁵, Mariana Bohns Michalowski⁶, Marcelo dos Santos Souza⁷, Gustavo Ribeiro Neves⁸, Luiz Fernando Lopes⁹, Sidnei Epelman¹⁰ and the *EMiLI study group*¹¹

1. Research Center, Coordenação de Pesquisa e Inovação, Instituto Nacional de Câncer, Rio de Janeiro, Brazil

2. Department of Pediatrics, Instituto Martagão Gesteira, Universidade Federal do Rio de Janeiro (UFRJ), Brazil

3. Coordenação de Prevenção -CONPREV, Instituto Nacional de Câncer, Rio de Janeiro, Brazil

4. Department of Hematology, Instituto Nacional de Infectologia Evandro Chagas, Fundação Oswaldo Cruz (Fiocruz), Rio de Janeiro, Brazil

5. Laboratory of Translational Research, Hospital da Criança de Brasília José Alencar, Brasília, Brazil

6. Department of Pediatric Hematology, Universidade Federal do Rio Grande do Sul (UFRGS), Porto Alegre, RS, Brazil; Hospital de Clínicas de Porto Alegre, RS, Brazil.

7. Department of Pediatric Hematology, Centro de Tratamento Oncológico e Hematológico Infantil - Hospital Regional Rosa Pedrossian (CETOHI-HRMS), Campo Grande, Mato Grosso do Sul, Brazil

8. Department of Pediatric Hematology, Hospital Sarina Rolin, São Paulo, Brazil

9. Children's Cancer Hospital, Barretos Cancer Hospital, Barretos, São Paulo, Brazil.
10. Department of Pediatric Oncology, Casa de Saúde Santa Marcelina, São Paulo, Brazil
11. EMI, **study group** (acronym Portuguese: *Estudo Multi-Institucional das Leucemias Infantis: Contribuição dos Marcadores Imuno-Moleculares na Distinção de Seus Subtipos e de Fatores de Riscos Etiopatogenicos*). The Collaborators names are provided in a separate document with a table with names and surnames and affiliations.

(*) Correspondence to: Maria S. Pombo-de-Oliveira M.D, Ph.D.

Instituto Nacional do Câncer, Rio de Janeiro, Brazil

Rua André Cavalcanti, 37, CEP 20231-050, Rio de Janeiro, Brazil

e-mail: mpombo@inca.gov.br; masocspoliveira@gmail.com

[Orcid ID: http://0000-0002-1507-004X](http://orcid.org/0000-0002-1507-004X)

Key-words: Childhood Leukemia; descriptive epidemiology, Brazil, socioeconomic indicators

Summary

Background The variation in geographical distribution and time trends of childhood acute leukaemia (c-AL) interests scientific and public health. The investigation herein provides a comparative regional distribution of c-AL, based on a large database representing five Brazil macroregion.

Methods This is a cross-sectional study in which secondary data from the EMI project was analyzed [period of 2000 to 2020]. Each regional age-specific incidence ratio (aSIR) up to 19 years was estimated. Demography and three socioeconomic indicators (human development index, regional growth development, and Gini coefficient) for each municipality were assessed. Poisson models, linear regression analysis, and the annual percentage change were calculated accordingly.

Findings 8073 c-AL cases were occurring in 1596 municipalities and treated in 65 centers with public exclusively assistance (97.6%), being, 5914 lymphoblastic (ALL, 65.7%) and 2159 myeloid (AML, 23.9%) leukaemia. The aSIR was significantly greater than 1.43 in municipalities with less than 50,000 population ($p=0.001$) but in the Midwest region. A significant excess of c-AL was observed in the North (aSIR, 2.11), South (aSIR 2.57), and Southeast (aSIR 2.61) regions where the coverage of the Family Healthy Strategy (FHS) program is consolidated. The overall cross-regional c-AL cases in municipalities of middle-high HDI level (>0.77) and Gini (0.47-0.56) living in rural lower-density populations were prevalent. A significant increase trending in ALL [3.10 (95% CI 1.28-4.95)] and AML [2.58 (95% CI 0.01-5.29)] were found.

Interpretation An increased c-AL incidence rate found in small municipalities with rural typology was associated with *family health system* coverage. Future studies are needed to evaluate the occurrence of high incidence rate of c-AL in some Brazilian regions.

Funding

The Instituto Nacional de Câncer supported this work; EMiLI had financial support from SwissBridge Foundation/Fundação do Câncer, Rio de Janeiro, Brazil grant (# L0J SWB2014 Sub-project 1.A), as well as from the TUCCA - Associação para Crianças e Adolescentes com Câncer (grant #00021); Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro (E-26/201.626/2021) and CNPq.

Maria S. Pombo de Oliveira is a scholar of CNPq (#310877/2019-9) and supported by Fundação Carlos Chagas Filho de Amparo a Pesquisa do Estado do Rio de Janeiro (E-26/202.758/2023).

Introduction

The variation in the incidence ratio and time trend of acute leukemia aroused interest in the scientific and public health communities worldwide¹. The extent of childhood leukemia burden and survival in many low-income and middle-income countries (LMICs) is still inaccurate, mainly because of a lack of robust information². Much of the published data focused on factual conditions that determine differences in outcome, such as treatment strategies and biomarkers adapted to regional socioeconomic conditions to achieve a good level of healing^{3–6}. A higher incidence of c-AL in South America (SA) than in Central America and the Caribbean countries (CACc) was recently reported in a comprehensive overview of geographical patterns and time trends generated by population-based cancer registries (PBCR)⁷. Nonetheless, the results are significantly impacted by regional socioeconomic factors, which are quite different from data from high-income countries (HIC)^{8,9}. Temporal trending, spatial distribution, biology, and childhood acute leukaemia (c-AL) mortality in SA and CACc deserve continued studies.

There are substantial articles on leukaemia studies in Brazil regarding biology and clinical treatment and outcome^{3,6,10,11}. Although, searching the literature, the spatial distribution of prevalent c-AL subtype's distribution across the country is still scarce. In a PBCR first-reported incidence of c-AL in children aged up to 14 years old, the median age-adjusted incidence rate (AAIR) for the whole country was 53.3 per million, like the AAIR found in HIC¹², however, inequalities in Brazil were identified and associated with low incidence rate and low human developed index (HDI) was observed¹². At that time, it was hypothesized that the deficit of cases ascertained would reflect the region registries' poor information and/or lack of access to health care^{2,12}. Meanwhile, access to health care and the systematization of health services for cancer across geographical regions, regardless of social inequalities, has improved¹³. The poor ascertainment of childhood leukemia using the

hematological disease classification in the current version of the International Classification of Diseases for Oncology (ICD-O) is an acknowledged problem in deprived regions¹⁴. Exploring the quality of access to health care, social indicators, and environmental exposures is relevant to establishing associated risk factors with c-AL¹⁵. Driven by this context, in the year 2000, a research group was built throughout a network of collaborations whose primary interest was to access the integrative diagnostic childhood leukaemias' (immunophenotype-genotype tests) under an umbrella of an interdisciplinary and multicentric network of clinical specialists and epidemiologists¹⁶. The physicians of the high-complexity oncology centres in different Brazilian cities provided epidemiological data along with biological samples to conduct epidemiological studies.

Then, a robust dataset of childhood leukemia information was built, allowing us to explore Brazil's regional distribution, diversities, and exciting findings that have gone unnoticed. The present report examines the regional distribution of c-AL in Brazil, the accessibility to provide integrative immune-molecular markers and the completeness of data that portrays the efforts to estimate the incidence rate of c-AL subtypes by Brazilian regions with socioeconomic indicators and access to good-quality diagnosis.

Methods

Study design, subjects, and data sources.

In this retrospective cross-sectional study, multicenter based, the secondary data was assessed to explore the demographic information obtained in the project *Multi-Institutional Study of Childhood Leukemia: contribution of immunomolecular markers in distinguishing their subtypes and etiology risk factors* (acronym EMiLI) from the years 2000 to 2020. Demography and clinical information were obtained through a request form containing clinical and demographic variables at the diagnosis demand. The selected variables for this

study are described in the supplement material (**Annexe 1**). The analyses were conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement¹⁷.

Childhood leukemia cases up to 19 years old were newly diagnosed and treated in 65 Cancer Centers scattered from the North, Northeast, Midwest, Southeast, and South of Brazil's macro-regions. The mandatory information was the child's demography (date of birth, sex, race), the city of treatment, and the municipality of residence with full address and postcode. For this analysis, the subject was categorized in a socioeconomic measure of area-based typology of the municipality where they were residents at the time of the diagnosis. Children born and/or living abroad were excluded from this analysis.

Characterization of Childhood Acute Leukemia

The initial diagnoses were established by the morphological description of peripheral blood and bone marrow aspiration, followed by immunophenotyping and cytogenetic tests¹⁶. Acute leukemia was classified as B-cell precursor ALL (Bcp-ALL), T-cell ALL (T-ALL), and acute myeloid leukemia (AML) subtypes according to the World Health Organization¹⁸. Diagnoses were grouped according to the ICD-O-3¹⁹. Once the acute leukemia diagnoses were performed by specially trained health personnel, the leftover patients' samples were processed for storage in a biorepository facility (EMiLi's repository biobank).

Variables Definition and Procedures

The exploratory variables were the leukemia subtypes (B-ALL, T-ALL, and AML); sex (male and female); the age strata were categorized into three groups: (1) less than four-year-old, (2) age 5 to 9, and (3) more than 10 to 19 years old at the c-AL diagnosis.

The races were categorized according to the mother's perception of child-index as Whites and Multiracial (defining as mixed-races of European descendants, original native population, and African descents) and following the international guidance recently reported²⁰. Population and area-based measures of urban/rural typology definition were obtained from the Brazilian Institute of Geography and Statistics (IBGE). Briefly, the typology of the child's living municipality was classified according to population size (both Census 2010 and 2020), and the characteristics of rural and urban spaces were defined using the conceptual matrix for rural-urban typology defined by each postal code of children's 'living address.' The degree of urbanization was then as follows: (1) Metropolis, predominantly urban with more than 1 million inhabitants in dense population areas and a level of urbanization superior to 75% of the population unit; (2) Large cities with more than 100 up to 999 x 10³ inhabitants with dense population area and high level of urbanization; (3) Medium cities with 50 to 100 x 10³ inhabitants and urbanization inferior to 75% of occupation and some level of rural activities; (4) Small cities, population unit less than 50 x 10³ inhabitants in occupation area less than 50% density and urbanization less than 75% with full rural activities (<https://www.ibge.gov.br/geociencias/>).

Classification of socioeconomic status and health service indicators

Three economic development indicators were assessed from the IBGE database: (1) Human Development Index (HDI) for each municipality, assessed considering the timing of the uptake and coverage expansion of the EMI for the period (<http://www.pnud.org.br/idh/>). HDI was measured into percentiles, such as very low HDI regions (<25th percentile), low HDI regions (>25th to 50th percentile, low-middle HDI region (>50th to 75th percentile), and middle-high HDI regions (>75th percentile). The deprivation level was assigned to the lower output area where patients were residents at the time of AL diagnosis (less than 50%

percentile). (2) The municipal gross domestic product (GDP) economic measure was calculated by dividing the total value of the GDP of the municipality by the number of resident inhabitants; (3) the Gini coefficient index was used to measure the income inequality to depict the Brazilian regional inequality as an indicator of health care distribution service. An increase in the Gini coefficient suggests that income is becoming more unevenly distributed in the target region (<https://www.ipea.gov.br/>). The estimated proportion of unit assistance in the Family Healthy Strategy units (FHSu) coverage was reviewed on the Ministry of Health platform as a *proxy* for primary health care access (<https://egestorab.saude.gov.br/>). The timing of the FHS uptake rate coverage in the period of this study was described recently²¹. The hospital that assisted the children was classified according to the current organization of Cancer Center Care accredited by Brazil's national health system (*Sistema Único de Saúde*, abbreviated as SUS) as (1) high complexity assistance Centers (in Portuguese *Centros de Assistência de Alta complexidade* - CACONS) and (2) high complexity assistance unit (in Portuguese, *Unidade de Assistência de Alta Complexidade*-UNACONS). Because of hybrid care assistance of private health insurance (25%) and Brazil's universal health system (SUS) assistance, the hospitals of treatment were also categorized as (1) public hospitals exclusively, (2) Philanthropic sustained with SUS and private health care, and (3) Private exclusively Hospital (<http://cnae.ibge.gov.br.>). To review any internal inconsistencies, a descriptive assessment of the total subjects and quality checks were performed to identify duplicates, use of non-existent ICD-O codes, loss of place of living, and missing variables.

Statistical Analysis

The chi-square test was initially used to compare demography frequency in contingency tables with demography variables, macroregion, and acute leukemia subtypes. Univariate *p*

values were calculated using Pearson's chi-square test with Yates' continuity correction or Fisher's exact test. Two-sided *p-values* with a significance limit of less than 0.05 were value considered throughout the analysis. To compare the difference in means assuming a 95% confidence interval (CI), a two-sample t-test or two-way ANOVA was performed after the Shapiro-Wilks test. The analysis of the regression models encompassed the significance t-test for the model parameters, ANOVA test for regression significance, and residual analysis, including the Komolgorov-Smirnov residual normality test. The incidence rates, sex rate ratios, and age-standardized rates were calculated by the Poisson regression model⁸. Age-specific incidence rates were computed for three distinct age brackets: 0 to 4 years, 5 to 9 years, and 10 to 19 years.

The extra-Poisson heterogeneity coefficient was conducted and described in detail in the **Supplemental material - Annexe 1**. Linear regression was performed to evaluate how socioeconomic indicators and health care, both qualitative and quantitative, influence the incidence rate. The following regression model: $E[y_i] = \alpha_0 + \alpha_1 D_i + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \beta_3 X_{i,3} + u_i$ where: considering that the coefficients of c-AL incidence rates were a dependent variable (y_i); D_i is a binary variable indicating whether the value of the predictor variable GR_urb (median) is below or above the median; $X_{i,1}$, $X_{i,2}$, $X_{i,3}$ denote the other explanatory variables: FHSu; Gini index; GDP *per capita* respectively. α_0 represents the value of the dependent variable (y) when all independent variables are set to zero; α_1/β represents the slope or change in the dependent variable; (y_i) corresponds to a one-unit change in the independent variable (x); Standardized Beta Coefficient (β), measured in standard deviation units measured provides a relative measure of the importance of each independent variable in predicting the variability of the dependent variable. Higher values of β indicate a proportionally more significant contribution of the corresponding variable to variations in the dependent variable. Admitting that the disturbances satisfy the usual hypotheses

[E(ui)]=0, it must be the average of the coefficient incidence for cases with a value below the median: $E[y_i/X_i, D_i = 0] = \alpha_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \beta_3 X_{i,3}$. The regression models were evaluated using significance tests such as the t-test for model parameters and the ANOVA test for regression significance. Residual analysis, including the Komolgorov-Smirnov normality test, was performed after applying the Variance Inflation Factor to detect multicollinearity, which was not found to be present in the data²². The Gini index within Brazil was used according to Lorenz curve plots (<https://databank.worldbank.org/>). For the overall HDI, the indicators were used according to the World Bank's data, which qualified Brazil as an upper-middle-income country during the period of the study²³. The annual percentage changes (APC) and the average annual percentage changes of each c-AL subtype were estimated for age strata and sex groups using the joinpoint regression analysis. The number of join points was obtained using a permutation test via the Monte Carlo method²⁴ by using the Joinpoint Regression Program, Version V4.8.0.1 (Statistical Research and Applications Branch, National Cancer Institute). The R-package and SPSS® software (versions 4.2.0 and 23.0, respectively) were employed for descriptive and regression analysis.

Role of the funding source

The funders of this study had no role in the study design, data collection, analysis, interpretation of the data, or writing of the manuscript.

Results

The EMI LI projects have accrued 8073 c-AL cases distributed from 1596 municipalities across the five Brazilian regions. Patients were treated in 65 cancer centers (CACON or UNACON) located in cities in which there are 17 active PBCRs, as shown in **Figures 1a**

and b. Cases were broadly representative in terms of age, sex, and urban or rural typology. The overall distribution of the significant major AL subtypes was 5914 ALL (65.7%) and 2159 AML (23.9%) cases. The childhood leukemia's health care assistance according to geographical distribution is shown in **Supplemental Table 1**. Most of the patients (n, 7885) had treatment care in UNACON centers located in Metropolis or Large cities of predominant urban typology with SUS exclusively assistance ($p=0.001$). For the code address nationwide localization assessment, the final data used was 7214 cases out of 8073 c-AL (89.3%). The proportion of 10.7% of cases was excluded because of the missing addresses, unclear identification, inconsistent leukemia subtype definition, and duplicates.

The frequency distribution of c-AL and socio-demographic variables assessed in five of Brazil's macroregions is detailed in **Table 1**. There is statistically significant heterogeneity for c-AL subtypes, race/ethnicity, typology of municipality, and type of health care assistance in the macroregion of Brazil ($p=0.001$). There is a significant increase in cases numerically observed in the period of 2011 to 2020, notably due to call cases in the North (threefold), Midwest, and South (twofold, respectively), compared to the previous decade ($p=0.001$).

Statistical differences were found in the proportional percentage distribution of Bcp-ALL (60.5% to 69.3%), T-ALL (8.3% to 12.3%), and AML (21.9% to 29.3%) in both periods ($p=0.001$). This difference in c-AL subtype distribution was observed mainly in the Northeast, Midwest, and Southeast regions, although the proportional percentage of ALL subtypes and AML remained similar across the region in both periods of the EMiLI studies. The details about each ALL and AML subtype are shown in **Supplemental Table 2**. No differences were found in the distribution of race, municipality, and place of living by typology. Both, ALL and AML occurred more frequently in males. The overall male-to-female ratio was

about 1.4:1.0 in all macroregions, but 1.7:1.0 in the North region, which was not statistically significant (p , 0.159).

Additional analysis in the distribution of Bcp-ALL, T-ALL, and AML by macroregions and race are shown in **Supplemental Table 3**. T-ALL is more frequent in the Northeast (12.3%), while the AML is in the Southeast and South (29.3% and 28.8%). The distribution of ALL and AML differed among White and Multiracial children ($p=0.001$). Amongst Multiracial children, there was an increased prevalence of T-ALL (27.2%) and AMLs (11.7%). In comparison, Bcp-ALL was 60.1% adjusted for age in the standard group ($p=0.001$). The proportion of Whites and Multiracial children reflects the heterogeneity of the Brazilian population diversity in each macroregion.

The Poisson test demonstrated a component of variability (**Table 2**) in the heterogeneity dispersion ($\beta > 1$) compared to what would be expected in the overall data of c-AL in the different regions by period, leukemia subtype, and age group. However, the statistical significance of heterogeneity or overdispersion was associated with the period of c-AL ($p=0.002$), mainly regarding AMLs increased in the period 2011-2020 ($p=0.007$), boys and girls, and age groups more than 5 years ($p=0.003$). The distribution of c-AL across the region was tested according to each macroregion's socioeconomic indicators profile. There was an association of percentage rates and the municipality's typology where high rates were found in urban, suburban, and growth areas.

The municipal population size and the typology of the place of living were also accessed, and cities with more than 100,000 inhabitants were prevalent in all regions (Metropolis and Large cities), variation was found in the percentage of c-AL in urban areas of all regions (27.4% to 39.8%). The increased proportion of c-AL cases in municipalities with less than 50,000 inhabitants was found in the Northeast and South regions ($p=0.001$). The AASIR data were estimated in 1596 Brazilian municipalities from the five macroregions. Overall,

the crude incidence rate (c-IR) was 55.9/10⁶ and 20.6/10⁶ for ALL and AML, respectively. In the first period (2000-2010) 43.1/10⁶ for ALL and 16.4/10⁶ for AML, while a substantial c-IR/per million in the period of 2011-2020, with ALL estimates of 66.3/10⁶ and for AML 22.4/10⁶ incidence rate (**Supplemental Table 4**).

The geographical variation of c-AL, ALL, and AML and the estimates of standardized incidence ratio were categorized according to the typology and population of the municipality in each macroregion (**Table 3**). For overall AL, ALL, and AML (period, boys and girls, ages 0-19 years), SIRs were significantly greater than 1.43 in municipalities with less than 50,000 population in all macroregions ($p=0.00$), but in the Midwest (SIR, 1.09). A significant excess of ALL (SIR, 2.11, SIR 2.57, and SIR 2.61) was observed in small municipalities from the North, South, and Southeast regions. The AML estimates were even higher than ALL, with SIRs significantly greater than 1.93 in all macroregions.

Age strata were tested as the definition, as shown in **Table 4**. We observed an aggregation of cases in the expected age peak of ALL (2-4 years) and in other age ranges (5 to 9 and 10 to 19 years), and AL subtypes were distributed in all macroregions. An increased risk association was observed in municipalities with less than 50,000 inhabitants of all ages and in all Brazilian macroregions. Regarding age at the c-AL diagnosis, as expected, a peak was observed in the 1- to 4-year age range among the ALLs for both boys and girls (**shown in Figure 2**). Interestingly, a second peak was observed in boys aged 14 to 16 years. The Bcp-ALL group showed the usual peak (63% aged up to 5 years, whereas the T-ALL cases were over five years (35%) ($p=0.001$).

The structure of the geographical and socioeconomic distribution of c-AL in different socioeconomic statuses and trending were also assessed. The mHDI was calculated for each macroregion (**Supplemental Table 5**). No c-AL case was found in the low percentile (<25th, mHDI <0.40), and very few c-AL cases were observed in lower-middle mHDI (25-50th,

mHDI=0.41-0.61). Most of the c-AL live in a municipality with upper-middle HDI (>75thmHDI = 0.67-0.77) and high mHDI (0.78-0.88).

Based upon the variation between Brazilian regions in socioeconomic development, access to health services, the density of municipality and the incidence of c-AL (subtypes, sex, race, and age group) were tested with a regression model (**Table 5**). Notably, the proportion of AL cases in less populated municipalities or regions of lower urbanity was observed across all macroregions, mainly from 2011 to 2020. The coverage of HFC in the geographic area disclosed by low urbanity (and rural) was increasing significantly and associated with the level of the Gini coefficient index ($p<0.001$). The regional distribution of the Gini coefficient with a high level of inequalities was observed in lower-density populations of all macroregions (mean, 0.47-0.56), which are areas of lower-middle mHDI. The statistically significant results demonstrate that there is no independence between the socioeconomic variables in the incidence of c-AL subtypes ($p<0.001$). The level of the HFS unit coverage and regional Gini coefficient index influence the incidence rate.

Finally, owing the substantial numbers of c-AL cases from 2011 to 2020, and being high percentage in the intermediate or rural environments, the APC was tested over time (**Figure 3**). A significant increase in the ALL [APC=3.10 (95% CI 1.28-4.95)] and AML [APC=2.58 (95% CI 0.01-5.29)] were found.

Discussion

This study is about searching information from an existing database of a nationwide collection of c-AL to explore questions other than biomarkers for which they were originally gathered¹⁶. This data comes from hospitals in cities with or without PBCR, as shown in Figure 1a-b, and the access is defined by birth location rather than place of treatment. The standardization of data and the quality control procedures in this study researched valuable

results with reliable interpretation because of information completeness and the nationwide coverage. The accuracy of the diagnosis is one of the most important procedures required in the quality control recommended by the International Agency for Research on Cancer (IARC) cancer incidence estimates. Hematological neoplasms, and especially acute leukemia, are diagnosed using multiple parameters (immunomolecular and cytogenetic), unlike other cancers based on tumor anatomy and histology. Unfortunately, the classifications adopted in PBCR information systems where the practice of gathering leukemias into groups without specific definitions tends to continue in SA and CACc. Therefore, the sample size, diagnostic accuracy, cases distribution, and similarities with previous PBCR reports indicate that the present data can be compared with many population-based studies conducted elsewhere^{25,26}.

There is significant regional heterogeneity for c-AL in Brazil, but overall, the distribution of the most common c-AL subtypes, Bcp-ALL, T-ALL, and AMLs, was proportionally scattered among all regions, as well as in line with previous PBCR data reported^{12,27}. As in many other countries, Bcp-ALL is the most common type of leukemia, showing a prevalent peak at ages 1-5 years and having little variation over time across different regions within the country. Compared with data from the United States and Mexico, where Bcp-ALL has been reported at 91.9% in the State of California and 73.2% in Mexico City, it appears that a deficit toward B-cell in some areas and an excess of AMLs^{1,28,29}. Few continental countries have descriptive epidemiology reports comparing patterns and trends for AL subtypes. Moreover, very few data from Brazil were included in these studies.

Specific age profiles in ALL are consistent across populations worldwide, although differences exist between ALL and AML regarding AASIR and outcome¹. The geographical distribution models assume that the population characteristics, such as collective identity (common language, race, and shared tradition) and race/ethnicity, reflect the genetic

predisposition and the association with exposure risk factors. For instance, the Latino population living in the United States has a higher incidence of ALL compared to African Americans, European Americans, and Asians^{30,31}. However, there are critical points in geographical or regional description in the context of race or ethnicity that should be addressed. Economic factors and deprivation or living conditions are intertwined with racial variables in SA and CAC populations.

Despite fluctuation of ascertainment cases in some regions in the first period of the study, the crude incidence rate was ALL, 55.9 per million, and AML, 20.6 per million and the c-AL rates increased according to mHDI (mean, >0.79). Very few c-AL cases were observed in remote areas of the region with low mHDI and good coverage of FHSu, and the lower incidence rates of ALL observed could be attributed to the protective mechanisms described as factors negatively associated with ALL³². The high occurrence of c-AL cases in municipalities with low population density (less than 50,000 inhabitants) within rural areas was unexpected. Overall, the statistical significance in the extra-Poisson component variance of about 56% between municipalities might be evidence of c-AL clustering in small-density areas of the country. These findings are also consistent with adequate healthcare services, including cancer diagnosis and treatment facilities. There was a substantial number of c-AL cases from 2011 to 2020, the period of expansion of the SUS¹³. The first interpretation of these results shall consider the improvement in access to specialized health services, the promotion of education and awareness about childhood cancer, and the support to move families from disadvantaged backgrounds to places with better treatment conditions. Nonetheless, when results were split according to regional social, demographic, and economic activities, it corroborates with the hypothesis that there might be a clustering of c-AL in the geo-localization of rural areas where agriculture is the major economic activity. Recently, Skidmore et al. have described in an empirical geospatial study, information from

two of Brazil's databases, that provided strong evidence of agricultural intensification of soy crop expansion associated with increased pediatric cancer and leukemia mortality³³. The authors also found a substantial statistically significant relationship between the soy area and the municipal level of production growing throughout the North region³⁴. A quantitative ecological analysis with the estimates of the population, fluctuation mobility, economic activity, and risk of childhood malignancies should be further explored^{33,35,36}.

To understand the present data, extended analyses were performed to explore whether improved access to healthcare services would be attributable to the increased number of cases in small municipalities in which c-AL were deemed to be unreported before^{2,12}. It is worthwhile to note that the SUS subsidizes all treatment, even among 25% of the Brazilian population that has private health insurance, and the SUS is much more effective in small and medium-sized cities as previously reported^{13,37}. The expansion and the better quality of primary health care have been observed as being associated with the implementation of the FHSu, which cover the first health care service for about 63% of the population²¹. There are 78 accredited pediatric oncohematology centers for cancer treatment in Brazil³⁹. Even with the distribution of CACON or UNACON higher concentrated in cities of the Southeast and South regions³⁷, the number of pediatric services that participated in this study was 65 out of 78 (83.3%) with representativeness of the cases incident in all Brazilian regions (**Figure 1a-b**). Differences in rates could either be due to differences in incidence or differences in ascertainment methods. These assumptions imply that data could be generated by PBCR or a consistent hospital-based registry because data are not encoded in the same way. The results here are validated by data previously reported with the analysis of PBCR¹².

Socioeconomic factors and access to healthcare can significantly impact the incidence rate and outcomes of c-AL. Efforts to address socioeconomic disparities in c-AL incidence and outcomes in middle-income countries involve a multifaceted approach, including improving

access to healthcare services, identifying environmental risks, and implementing public health policies able to reduce the disease³⁸. Therefore, the investigation herein is based on location at birth, to provide comparative variations and time trends of childhood leukemia in Brazil.

The substantial diversity of genetic and environmental conditions observed in the Brazilian population, including population displacement within the country, changing family patterns including size and lifestyle profiles, and new occupational and environmental exposures, among other variables, creates a special framework allowing the exploration of the natural history of cancer in general, and of childhood leukemia in particular. The differences observed might be associated with environmental exposures and genetic background. However, the results should be interpreted cautiously. The major challenge of causal studies, which is not considered herein, is the complexity of the combination of multi-environment adverse factors such as pesticides, air pollution, smoking, alcohol, and infections, which might interplay a synergistic effect on c-AL causality³⁹. The childhood death of ALL was attributed to access to a long-distance pediatric cancer center.

Limitations and Strengths

It is well-recognized that retrospective studies have limitations. Herein, we point out the long period (two decades) of the cases' ascertainment, the discontinuity of some participating centers, the loss of address codes as it was predetermined (about 8% of losses), as well as the loss of information due to patients' mobility between treatment centers that could add errors. However, when we compare our data with those disclosed in the national hospital-based registry [DataSUS, <http://tabnet.datasus.gov.br>], it is estimated that the information collected in the EMiLI project represents about 80% of the cases incident in the period in each region included here.

Even with some limitations, the strengths of this study are the sample size and the national representativeness with code addresses of cases' places of living, including the distribution analysis of c-AL subtypes in areas where PBCRs do not reach. The large number of cancer centers, patients included, and the wide time frame covered enable us to have a comprehensive historical point of view on the topic for further validation and ecological studies. The network created with clinical and academic purposes, rather than originating with an official vertical bureaucratic structure, has resulted in a high degree of voluntary ascertainment cases in well-representative regions, providing conditions favoring the exploration of the natural history of c-ALL in Brazil. One of the major strengths of the EMiLI network was the multidisciplinary condition to investigate the early-age c-AL, such as infant leukemia-a rare subtype with short latent period between exposure and the overt disease in which several case-control studies were performed^{27,40-42}. The integrative line of healthcare institutions and regional cancer centers was strengthened, especially in areas with poor accessibility to tertiary center care.

Conclusion. This study provides evidence of regional differences in the incidence of c-AL in Brazil, quantifiable evidence of the access to health care services and high-quality diagnosis of c-AL across different Brazilian regions. A significant increase in ALL and AML were observed between 2011 to 2020 that could be partially explained by the coverage of the FHS program and the access to oncological center care. The high occurrence of c-AL and excess of AML in municipalities with rural topology should be investigated.

Contributors

MSPO: Developed the study concept, data curation, formal analysis, funding acquisition, writing—the original draft, and writing—the review and editing.

MGPL: Directed the statistical methods and writing, reviewed the manuscript, and provided professional input into the interpretation and editing of the results.

PCN: Performed the methodology used to statistical analyze and prepare the results in tables and figures, investigation, and data curation.

DPMA: Provided professional guidance, had full access to data analysis and investigation, and verified the literature review and editing.

IMQM: provided case information, discussion, data curation, and professional guidance on childhood leukemia.

MBM: provided cases information, data curation, input into interpretation, read the draft, and professional guidance on childhood leukemia.

MSS: provided additional cases information, data curation, and professional guidance on childhood leukemia.

GRN: provided additional cases information, data curation, and professional guidance on childhood leukemia.

LFL: provided case information, discussion, data curation, and professional guidance on childhood leukemia.

SE: Provided case information, discussion, input into interpretation, data curation, funding acquisition, and professional guidance on childhood leukemia.

All co-authors in the EMiLI group have provided information, read the report and approved the final version before submission of the manuscript.

Declaration of interest

No conflict of interest

Data sharing statement

The information datasets generated and analysed during the current study are available from the corresponding author upon reasonable request.

Acknowledgments

508 We thank all physicians who have sent patients samples, clinical information and have given
509 support to the EMiLI Studies: Adrezza Bastos De Mattos, Adriana Karin Sapuppo, Adriana
510 Martins de Souza, Alejandro Mauricio Arancibia, Alexandre Gustavo Apa, Amanda Straub
511 Braga, Ana Paula S. Bueno, Andrea Gadelha Nobrega, Andrea Pandolfi Barcellos, Anelisa
512 Schittini Costa, Atalla Mnayarji, Bianca Martins Biancovilli, Bruno Terra Correa, Bruno Vieira
513 Dias, Camila Peixoto Pereira de Souza, Carmem Maria Costa Mendonca Fiori, Carolina Pietro
514 Cardoso, Caroline Fincatto da Silva, Cassia Silvestre Mariano Pires, Claudia Julia Bruzzi Porto
515 Coelho, Cristiane de Sa Pinto, Cristina Rodriguez de Carvalho, Cynthia Curvello, Dora Marcia
516 Alencar, Elaine Barbosa Silverio Tavares, Elaine Sobral da Costa, Erica Boldrini, Everaldo
517 Ruiz Junior, Fabia Idalina Rodrigues Neves, Fabricio de Oliveira Correia, Fernanda C. Jordy
518 Macedo, Fernando de Almeida Werneck, Flavia Machado Nogueira, Flavia Nogueira Serafim
519 Araujo, Flavio Ferreira Andrade, Gilberto Ramos, Gildene Alves da Costa, Glaceanne Torres
520 da Luz Mamede, Jane de Almeida Dobbin, Ilana Zalcebeg Renault, Jozina Maria Andrade
521 Agareno, Karita Cristina Naves Corbellini, Larissa Bueno Polis Moreira, Leandra Paiva
522 Queiroz, Lilian Maria Burlacchini de Carvalho, Lilian Maria Cristofani, Luana Novaes Bonfim,
523 Luciana Nunes Silva, Mara Albonei Dudeque Pianovski, Marcela de Moraes Barros e Sousa,
524 Marcia T. Schramm, Maria Dolores Dorea, Maria do Perpetuo Socorro Sampaio Carvalho,
525 Maria Lydia D' Andrea, Mariana Emerenciano, Mauricio de Souza Meira, Melissa Ferreira de
526 Macedo, Monica Lankszner, Nilma Pimentel de Brito, Nubia Mendonça, Pablo Santiago,
527 Patricia Carla de Lima, Patricia Vieira dos Santos, Rebeca Ferreira Marques, Regiana Quinto
528 de Souza, Renata Pereira de Sousa Barros, Renata Sarkis Alves, Renata Silva de Carvalho
529 Gurgel, Renato Melaragno, Rosania Maria Basegio, Sandra Helena De Oliveira Estelita, Silvia
530 Rocha Caetano do Nascimento Agra, Soraia Taveira Rouxinol, Suely Santos Araujo, Tatiana
531 Soria Rosa, Thais Alcantara Bonilha, Tiago Hessel Tormen, Virginia Maria Coser, Vitoria
532 Regia Pereira Pinheiro, Wellington Mendes, Vanessa Carvalho Lima.

533

534 **Annexe:** Supplementary data. Supplementary information includes (1) a summary translated
535 into the Portuguese language, (2) a description of the study design and statistical detail
536 models, a Strobe statement, and six complementary tables.

537 **Research Ethics.** This EMI LI study was approved by the Brazilian National Institute of
538 Cancer of Ethics and Research Committee, and the centre's ethics committee approved this
539 study CEP/CONEP:1.394.043. Local agreements or approvals were not necessary for this
540 research project since the secondary dataset for analysis did not identify the subjects.

541

542 References

- 543 1 Steliarova-Foucher E, Fidler MM, Colombet M, *et al.* Changing geographical patterns
544 and trends in cancer incidence in children and adolescents in Europe, 1991-2010
545 (Automated Childhood Cancer Information System): a population-based study. *Lancet*
546 *Oncol* 2018; **19**: 1159–69.
- 547 2 Azevedo-Silva F, Reis R de S, Santos M de O, Luiz RR, Pombo-de-Oliveira MS.
548 Evaluation of childhood acute leukemia incidence and underreporting in Brazil by
549 capture-recapture methodology. *Cancer Epidemiol* 2009; **33**: 403–5.
- 550 3 Brandalise SR, Pinheiro VR, Aguiar SS, *et al.* Benefits of the intermittent use of 6-
551 mercaptopurine and methotrexate in maintenance treatment for low-risk acute
552 lymphoblastic leukemia in children: randomized trial from the Brazilian Childhood
553 Cooperative Group--protocol ALL-99. *J Clin Oncol* 2010; **28**: 1911–8.
- 554 4 Howard SC, Davidson A, Luna-Fineman S, *et al.* A framework to develop adapted
555 treatment regimens to manage pediatric cancer in low- and middle-income countries:
556 The Pediatric Oncology in Developing Countries (PODC) Committee of the
557 International Pediatric Oncology Society (SIOP). *Pediatr Blood Cancer* 2017; **64 Suppl**
558 **5**. DOI:10.1002/pbc.26879.
- 559 5 Cecconello DK, Rechenmacher C, Werlang I, *et al.* Implementation of the asparaginase
560 activity assessment technique for clinical use: experience of a Brazilian Center. *Sci Rep*
561 2020; **10**: 21481.
- 562 6 Bonilha TA, Obadia DDA, Valveson AC, Land MGP. Outcome of childhood acute
563 lymphoblastic leukemia treatment in a single center in Brazil: A survival analysis study.
564 *Cancer Rep (Hoboken)* 2022; **5**: e1452.
- 565 7 de Paula Silva N, Colombet M, Moreno F, *et al.* Incidence of childhood cancer in Latin
566 America and the Caribbean: coverage, patterns, and time trends. *Rev Panam Salud*
567 *Publica* 2024; **48**: e11.

- 568 8 Wang L, Gomez SL, Yasui Y. Racial and Ethnic Differences in Socioeconomic Position
569 and Risk of Childhood Acute Lymphoblastic Leukemia. *Am J Epidemiol* 2017; **185**:
570 1263–71.
- 571 9 Ni X, Li Z, Li X, *et al.* Socioeconomic inequalities in cancer incidence and access to
572 health services among children and adolescents in China: a cross-sectional study. *Lancet*
573 2022; **400**: 1020–32.
- 574 10 Brandalise SR, Viana MB, Pinheiro VRP, *et al.* Shorter Maintenance Therapy in
575 Childhood Acute Lymphoblastic Leukemia: The Experience of the Prospective,
576 Randomized Brazilian GBTLI ALL-93 Protocol. *Front Pediatr* 2016; **4**: 110.
- 577 11 Andrade FG, Feliciano SVM, Sardou-Cezar I, *et al.* Pediatric Acute Promyelocytic
578 Leukemia: Epidemiology, Molecular Features, and Importance of GST-Theta 1 in
579 Chemotherapy Response and Outcome. *Front Oncol* 2021; **11**: 642744.
- 580 12 de Souza Reis R, de Camargo B, de Oliveira Santos M, de Oliveira JM, Azevedo Silva
581 F, Pombo-de-Oliveira MS. Childhood leukemia incidence in Brazil according to
582 different geographical regions. *Pediatr Blood Cancer* 2011; **56**: 58–64.
- 583 13 Castro MC, Massuda A, Almeida G, *et al.* Brazil's unified health system: the first 30
584 years and prospects for the future. *Lancet* 2019; **394**: 345–56.
- 585 14 Smith A, Howell D, Patmore R, Jack A, Roman E. Incidence of haematological
586 malignancy by sub-type: a report from the Haematological Malignancy Research
587 Network. *Br J Cancer* 2011; **105**: 1684–92.
- 588 15 Schüz J, Erdmann F. Environmental Exposure and Risk of Childhood Leukemia: An
589 Overview. *Arch Med Res* 2016; **47**: 607–14.
- 590 16 Pombo de Oliveira MS, Koifman S, Vasconcelos GM, Emerenciano M, de Oliveira
591 Novaes C, Brazilian Collaborative Study Group of Infant Acute Leukemia.
592 Development and perspective of current Brazilian studies on the epidemiology of
593 childhood leukemia. *Blood Cells Mol Dis* 2009; **42**: 121–5.
- 594 17 von Elm E, Altman DG, Egger M, *et al.* The Strengthening the Reporting of
595 Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting
596 observational studies. *J Clin Epidemiol* 2008; **61**: 344–9.
- 597 18 Arber DA, Orazi A, Hasserjian RP, *et al.* International Consensus Classification of
598 Myeloid Neoplasms and Acute Leukemias: integrating morphologic, clinical, and
599 genomic data. *Blood* 2022; **140**: 1200–28.
- 600 19 Steliarova-Foucher E, Stiller C, Lacour B, Kaatsch P. International Classification of
601 Childhood Cancer, third edition. *Cancer* 2005; **103**: 1457–67.
- 602 20 Flanagan A, Frey T, Christiansen SL, AMA Manual of Style Committee. Updated
603 Guidance on the Reporting of Race and Ethnicity in Medical and Science Journals.
604 *JAMA* 2021; **326**: 621–7.
- 605 21 Andrade MV, Coelho AQ, Xavier Neto M, de Carvalho LR, Atun R, Castro MC.
606 Transition to universal primary health care coverage in Brazil: Analysis of uptake and

607 expansion patterns of Brazil's Family Health Strategy (1998-2012). *PLoS One* 2018; **13**:
608 e0201723.

609 22 Sousa A, Dal Poz MR, Carvalho CL. Monitoring inequalities in the health workforce:
610 the case study of Brazil 1991-2005. *PLoS One* 2012; **7**: e33399.

611 23 United Nations Development Programme. Human Development Index trends, 1990–
612 2019. 2020. <https://hdr.undp.org/system/files/documents/2020statisticalannextable2.pdf>
613 (accessed July 21, 2024).

614 24 Kim HJ, Fay MP, Feuer EJ, Midthune DN. Permutation tests for joinpoint regression
615 with applications to cancer rates. *Stat Med* 2000; **19**: 335–51.

616 25 Bray F, Parkin DM. Evaluation of data quality in the cancer registry: principles and
617 methods. Part I: comparability, validity and timeliness. *Eur J Cancer* 2009; **45**: 747–55.

618 26 Parkin DM, Bray F. Evaluation of data quality in the cancer registry: principles and
619 methods Part II. Completeness. *Eur J Cancer* 2009; **45**: 756–64.

620 27 Reis R de S, Silva N de P, Santos M de O, *et al.* Mother and child characteristics at birth
621 and early age leukemia: a case-cohort population-based study. *J Pediatr (Rio J)* 2017;
622 **93**: 610–8.

623 28 Pérez-Saldivar ML, Fajardo-Gutiérrez A, Bernáldez-Ríos R, *et al.* Childhood acute
624 leukemias are frequent in Mexico City: descriptive epidemiology. *BMC Cancer* 2011;
625 **11**: 355.

626 29 Barrington-Trimis JL, Cockburn M, Metayer C, Gauderman WJ, Wiemels J, McKean-
627 Cowdin R. Trends in childhood leukemia incidence over two decades from 1992 to
628 2013. *Int J Cancer* 2017; **140**: 1000–8.

629 30 Lim JY-S, Bhatia S, Robison LL, Yang JJ. Genomics of racial and ethnic disparities in
630 childhood acute lymphoblastic leukemia. *Cancer* 2014; **120**: 955–62.

631 31 Puumala SE, Ross JA, Aplenc R, Spector LG. Epidemiology of childhood acute
632 myeloid leukemia. *Pediatr Blood Cancer* 2013; **60**: 728–33.

633 32 Ribeiro KB, Buffler PA, Metayer C. Socioeconomic status and childhood acute
634 lymphocytic leukemia incidence in São Paulo, Brazil. *Int J Cancer* 2008; **123**: 1907–12.

635 33 Skidmore ME, Sims KM, Gibbs HK. Agricultural intensification and childhood cancer
636 in Brazil. *Proc Natl Acad Sci U S A* 2023; **120**: e2306003120.

637 34 Thuler LCS, Pombo-de-Oliveira MS. Acute promyelocytic leukaemia is highly frequent
638 among acute myeloid leukaemias in Brazil: a hospital-based cancer registry study from
639 2001 to 2012. *Ann Hematol* 2017; **96**: 355–62.

640 35 Alexander FE, Boyle P, Carli PM, *et al.* Spatial clustering of childhood leukaemia:
641 summary results from the EUROCLUS project. *Br J Cancer* 1998; **77**: 818–24.

- 642 36 Alexander FE, Boyle P, Carli PM, *et al.* Spatial temporal patterns in childhood
643 leukaemia: further evidence for an infectious origin. EUROCLUS project. *Br J Cancer*
644 1998; **77**: 812–7.
- 645 37 da Silva MJS, O'Dwyer G, Osorio-de-Castro CGS. Cancer care in Brazil: structure and
646 geographical distribution. *BMC Cancer* 2019; **19**: 987.
- 647 38 Lortet-Tieulent J, Georges D, Bray F, Vaccarella S. Profiling global cancer incidence
648 and mortality by socioeconomic development. *Int J Cancer* 2020; **147**: 3029–36.
- 649 39 Onyije FM, Olsson A, Baaken D, *et al.* Environmental Risk Factors for Childhood
650 Acute Lymphoblastic Leukemia: An Umbrella Review. *Cancers (Basel)* 2022; **14**: 382.
- 651 40 Pombo-de-Oliveira MS, EMiLI Study Group, Petridou ET, *et al.* The Interplay of
652 Cesarean-Section Delivery and First-Birth Order as Risk Factors in Acute
653 Lymphoblastic Leukemia. *Cancer Epidemiol Biomarkers Prev* 2023; **32**: 371–9.
- 654 41 Pombo-de-Oliveira MS, Andrade FG, Brazilian Collaborative Study Group of Infant
655 Acute Leukaemia. Early-age Acute Leukemia: Revisiting Two Decades of the Brazilian
656 Collaborative Study Group. *Arch Med Res* 2016; **47**: 593–606.
- 657 42 Ferreira JD, Couto AC, Pombo-de-Oliveira MS, Koifman S, Brazilian Collaborative
658 Study Group of Infant Acute Leukemia. In utero pesticide exposure and leukemia in
659 Brazilian children < 2 years of age. *Environ Health Perspect* 2013; **121**: 269–75.
- 660

Table 2. Standardized Incidence Rate Heterogeneity coefficient of childhood leukemia according to leukemia subtypes sex and age, Brazil, 2000-2020.

		Y_i	β^e	P^e value	Range (SIR) ^Ω
Overall Acute Leukemia		7140	1.022	0.002	0.72-1.85
Period	2000-2010	3122	1.070	0.020	0.61-4.32
	2011-2020	4018	1.012	0.017	1.13-2.75
Sex	Boys	4171	1.030	0.229	0.70-2.45
	Girls	2969	1.032	0.467	0.57-2.95
Age (years)	0-4	2838	1.010	0.321	0.54-3.23
	5-9	1924	1.017	0.084	0.42-3.69
	10-19	2378	1.020	0.022	0.59-3.40
Lymphoblastic Leukemia		5215	1.027	0.053	0.61-2.31
Period	2000-2010	2262	1.080	0.050	0.61-4.32
	2011-2020	2953	1.015	0.157	0.91-5.95
Sex	Boys	3095	1.012	0.384	0.49-3.02
	Girls	2120	1.019	0.318	0.58-4.41
Age (years)	0-4	2074	1.078	0.205	0.57-2.32
	5-9	1504	1.120	0.067	0.58-2.13
	10-19	1637	1.094	0.075	0.61-2.21
Myeloblastic Leukemia		1925	1.025	0.003	0.53-4.73
Period	2000-2010	860	1.046	0.077	0.55-5.18
	2011-2020	1065	1.056	0.007	1.01-9.76
Sex	Boys	1075	1.210	0.000	0.72-8.26
	Girls	849	1.056	0.021	0.35-7.78
Age (years)	0-4	764	1.066	0.104	0.64-4.98
	5-9	420	1.081	0.003	0.17-4.51
	10-19	741	1.077	0.006	0.47-3.87

Abbreviations: Y_i observed number of incident cases; β^e magnitude power of variability component; ^Ω Range(SIR) com (SIR)_i= y_i/E_i ; [£]Estimate of extra Poisson component of variability(%); ^eP-value < 0.05.

Figure 1. The distribution of childhood acute leukemia cases diagnosed in EMiLI study and the socioeconomic indicators, Brazil, 2000-2020

a. North and Northeast

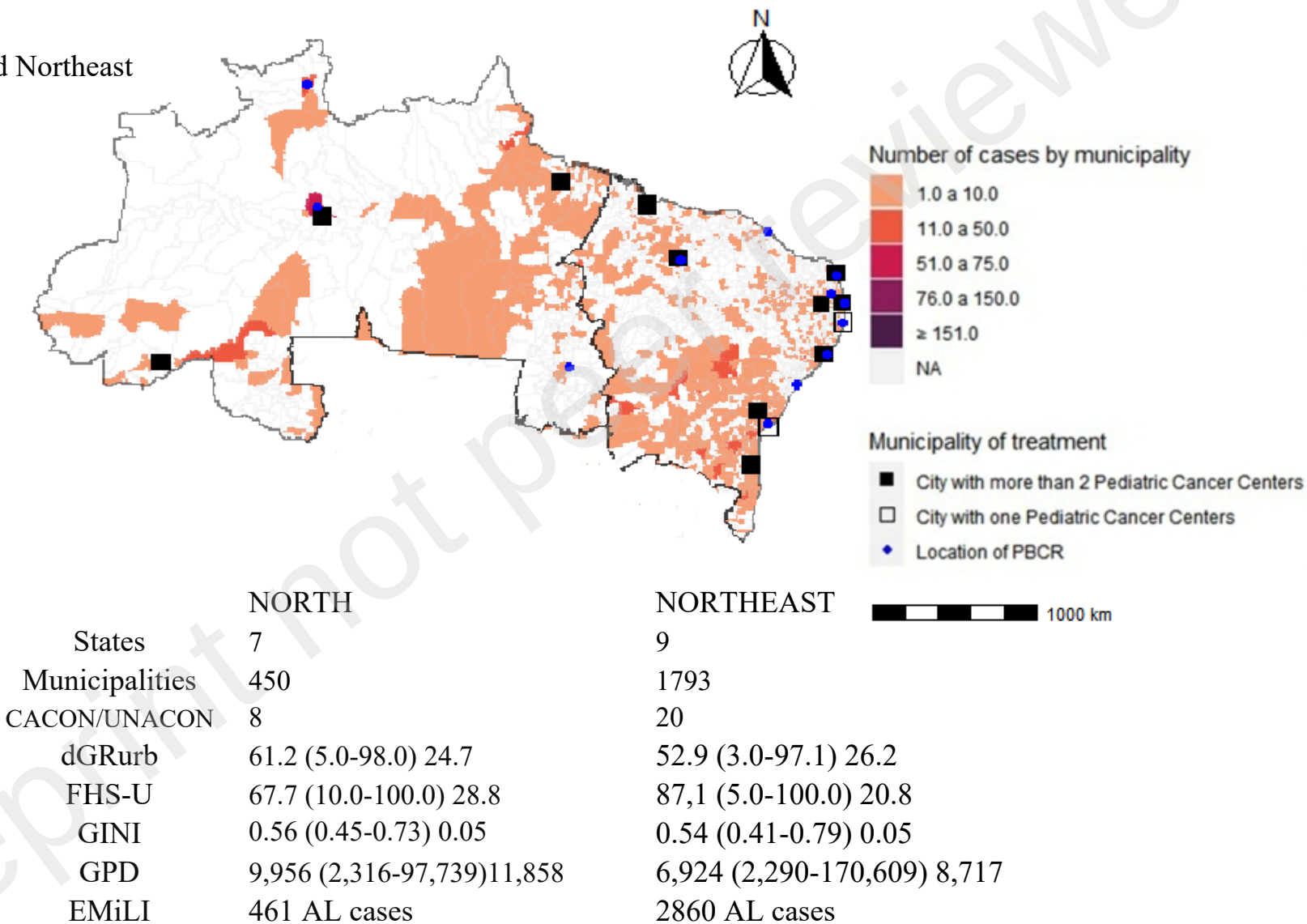
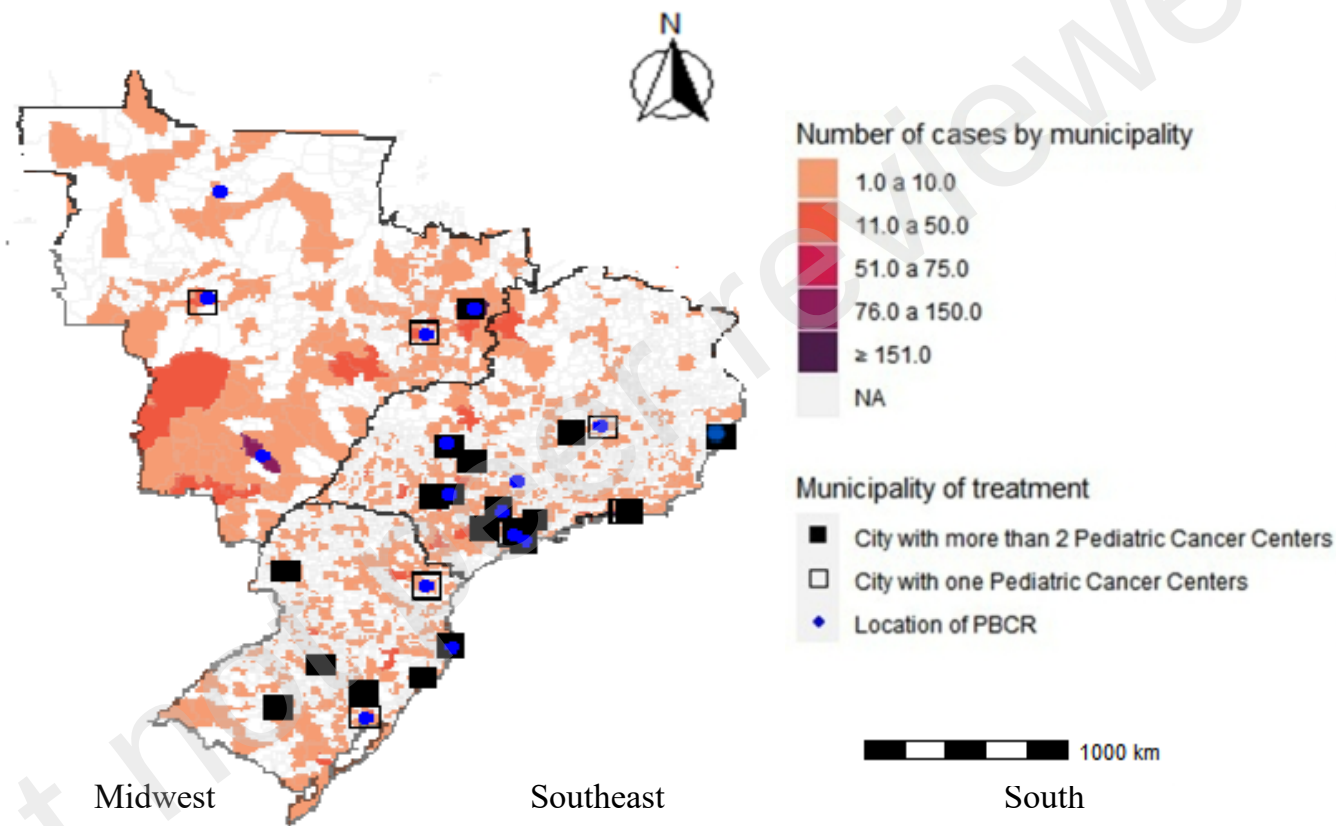


Figure 1.

b. Midwest, Southeast and South



	Midwest	Southeast	South
States	4	5	3
Municipalities	467	1668	1191
CACON/UNACON	8	35	18
dGRurb	71.2 (5.0-97.7)23.4	81.0 (10.0-98.6) 21.2	64.9(5.0-97.1)31.2
FHS-U	82.0 (13.8-100.0)21.1	65.3 (3.0-100.0) 28.2	72.9(10.0-10.00)26.9
GINI	0.52(0.41-0.79) 0.05	0.49 (0.36-0.69) 0.05	0.47(0.33-0.64)0.06
GPD	15,420(3,417-59,831)9,188	(3,815-312,257) 22,206	19,315(3,976-158,795)13,929
EMiLI	1361 AL cases	2860 AL cases	860 AL cases

Figure 2 Age-sex distribution of childhood leukemia, Brazil, 2000-2020

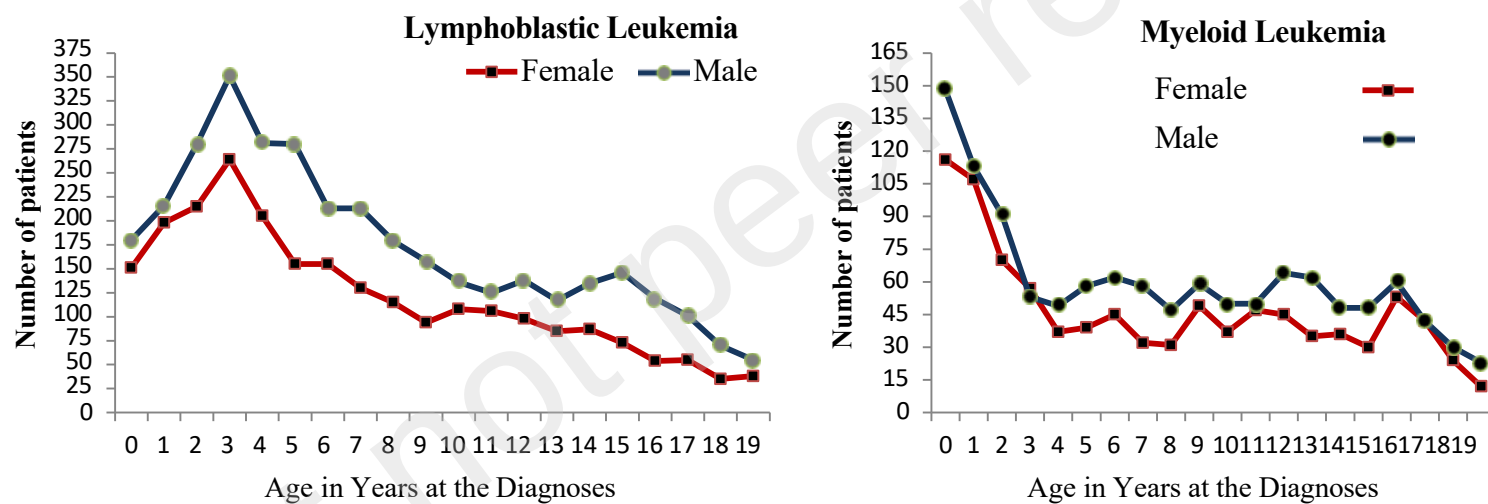


Figure 3. The annual percentage change of childhood leukemia in Brazil from 2000 to 2020.

